Université Djillali Liabes
Faculté de Technologie
Section Génie Electrique
Matière: Logique Combinatoire et séquentielle

2019 / 2020

TD N° 2
Simplification des fonctions logiques

Exercice 1 :

Donner la table de vérité des fonctions suivantes :

$f_1 = a.b + \overline{a}.\overline{c}$

$f_2 = \overline{a}.c + a.b.\overline{c}$

Exercice 2 :

Donner l'expression algébrique de F

a	b	c	d	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Exercice 3 :

Simplifier algébriquement les équations suivantes.

$f_1 = a + \overline{a}.b$

$f_2 = \overline{a} + a.b$

$f_3 = b + a.\overline{b}.c$

$f_4 = \overline{a}\,\overline{b}\,\overline{c} + \overline{a}\,\overline{b}c + \overline{a}bc$

$f_5 = \overline{a}b\overline{c} + abc + ab\overline{c}$

$f_6 = \overline{a}bc + a\overline{b}c + ab\overline{c} + abc$

$f_7 = \overline{a}\,\overline{b}\,\overline{c}\,\overline{d} + \overline{a}\,\overline{b}c\overline{d} + a\overline{b}c\overline{d} + abc\overline{d} + \overline{a}\,\overline{b}\,\overline{c}d + \overline{a}bc\overline{d}$

$f_8 = \overline{a}\,\overline{b}\,\overline{c}\,\overline{d} + \overline{a}\,\overline{b}\,\overline{c}d + \overline{a}\,\overline{b}cd + a\overline{b}\,\overline{c}\,\overline{d} + a\overline{b}\,\overline{c}d$

Exercice 4 :

Simplifier par la méthode graphique (Karnaugh) les fonctions suivantes :

$F1 = A.\overline{B}.\overline{C} + \overline{A}.B.C + \overline{A}.\overline{B}.C$

$F2 = \overline{A}.\overline{B}.C + \overline{A}.B.C + A.B.C$

$F3 = \overline{A}.\overline{B}.\overline{C} + A.\overline{B}.\overline{C} + \overline{A}.B.C + A.B.C + A.B.\overline{C}$

$F4 = A.\overline{B}.\overline{C} + \overline{A}.B.C + \overline{A}.\overline{B}.C$ et interdit $A.B.\overline{C}, A.B.C$

$F5 = \overline{A}.\overline{B}.\overline{C}.\overline{D} + \overline{A}.\overline{B}.C.\overline{D} + \overline{A}.B.\overline{C}.D + \overline{A}.B.C.\overline{D} + A.B.\overline{C}.D + A.B.C.\overline{D} + A.\overline{B}.\overline{C}.D + A.\overline{B}.C.D$

$F6 = \overline{A}.\overline{B}.C.\overline{D} + \overline{A}.B.\overline{C}.\overline{D} + \overline{A}.B.\overline{C}.D + \overline{A}.B.C.D + \overline{A}.B.C.\overline{D} + A.B.\overline{C}.\overline{D} + A.\overline{B}.\overline{C}.D + A.B.\overline{C}.D$

Exercice 5 :

Transformer les équations ci-dessous pour n'avoir que des $NAND_2$. Donner ensuite le logigramme de ces fonctions.

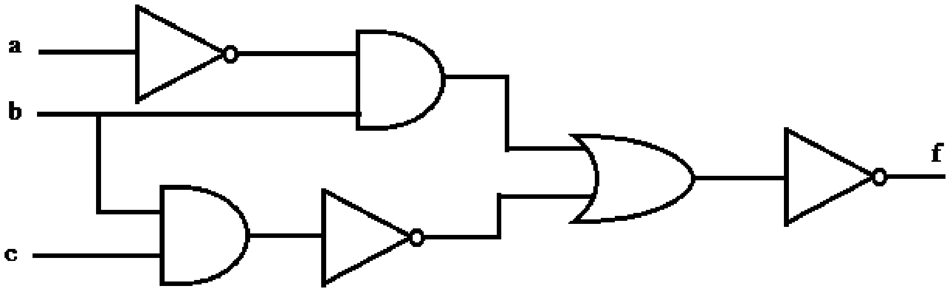
$S_1 = a\overline{b} + c$

$S_2 = (a + b).c$

$S_3 = \overline{ac} + a\overline{b}$

Exercice 6 :

Le résultat d'une étude donne le logigramme suivant. Retrouver l'expression algébrique de F et simplifier la si possible.



Solution du TD N°2

Exercice 1: Donner la table de vérité de f_1: $f1 = a.b + \overline{a}.\overline{c}$

a	b	c	a.b	$\overline{a}.\overline{c}$	F_1
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	0	1

$$f_2 = \overline{a}.c + a.b.\overline{c}$$

a	b	c	F_2
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Exercice 2 : Donner l'expression algébrique de F.

$$F = \overline{a}\overline{b}\overline{c}\overline{d} + \overline{a}\overline{b}\overline{c}d + \overline{a}\overline{b}cd + \overline{a}b\overline{c}d + \overline{a}bc\overline{d} + \overline{a}bcd + a\overline{b}cd + ab\overline{c}d + abc\overline{d}$$

Exercice 3 : Simplifier algébriquement les équations suivantes.

$$f1 = a + \overline{a}.b$$
$$f1 = (a + \overline{a}).(a + b)$$
$$f1 = 1.(a + b)$$
$$f1 = a + b$$

$$f2 = \overline{a} + a.b$$
$$f2 = (\overline{a} + a).(\overline{a} + b)$$
$$f2 = 1.(\overline{a} + b)$$
$$f2 = \overline{a} + b$$

$$f3 = b + a.\overline{b}.c$$
$$f3 = (b + \overline{b}).(b + a.c)$$
$$f3 = 1.(b + a.c)$$
$$f3 = b + ac$$

$$f4 = \overline{a}\overline{b}\overline{c} + \overline{a}\overline{b}c + \overline{a}bc$$
$$f4 = \overline{a}\overline{b}.(\overline{c} + c) + \overline{a}bc$$
$$f4 = \overline{a}\overline{b}.1 + \overline{a}bc$$
$$f4 = \overline{a}\overline{b} + \overline{a}bc$$
$$f4 = \overline{a}.(\overline{b} + bc)$$
$$f4 = \overline{a}.((\overline{b} + b).(\overline{b} + c))$$
$$f4 = \overline{a}.(\overline{b} + c) = \overline{a}\overline{b} + \overline{a}\,c$$

$$f5 = \overline{a}b\overline{c} + abc + ab\overline{c}$$
$$f5 = b\overline{c}.(\overline{a} + a) + abc$$
$$f5 = b\overline{c} + abc$$
$$f5 = b.(\overline{c} + ac)$$
$$f5 = b.((\overline{c} + a).(\overline{c} + c))$$
$$f5 = b.(\overline{c} + a) = b\overline{c} + ab$$

$f6 = \bar{a}bc + a\bar{b}c + ab\bar{c} + abc$

$f6 = \bar{a}bc + a\bar{b}c + ab\bar{c} + abc + abc + abc$

$f6 = ab(c + \bar{c}) + ac(b + \bar{b}) + bc(a + \bar{a})$

$f6 = ab + ac + bc$

$f7 = \bar{a}\bar{b}\bar{c}\bar{d} + \bar{a}\bar{b}c\bar{d} + a\bar{b}c\bar{d} + abc\bar{d} + a\bar{b}\bar{c}\bar{d} + \bar{a}bc\bar{d}$

$f7 = \bar{a}\bar{b}\bar{d}(\bar{c} + c) + a\bar{b}\bar{d}(\bar{c} + c) + bc\bar{d}(\bar{a} + a)$

$f7 = \bar{a}\bar{b}\bar{d} + a\bar{b}\bar{d} + bc\bar{d}$

$f7 = \bar{b}\bar{d}(a + \bar{a}) + bc\bar{d}$

$f7 = \bar{d}(\bar{b} + bc)$

$f7 = \bar{d}((\bar{b} + b).(\bar{b} + c))$

$f7 = \bar{d}(\bar{b} + c) = \bar{b}\bar{d} + c\bar{d}$

$f8 = \bar{a}\bar{b}\bar{c}\bar{d} + \bar{a}\bar{b}\bar{c}d + \bar{a}\bar{b}cd + a\bar{b}\bar{c}\bar{d} + a\bar{b}\bar{c}d + \bar{a}\bar{b}\bar{c}d$

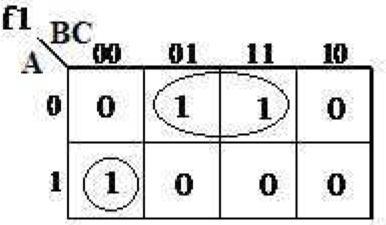
$f8 = \bar{b}\bar{c}(\bar{a}\bar{d} + \bar{a}d + a\bar{d} + ad) + \bar{a}\bar{b}cd + \bar{a}\bar{b}\bar{c}d$

$f8 = \bar{b}\bar{c}((\bar{a}(\bar{d} + d) + a(\bar{d} + d)) + \bar{a}\bar{b}d(c + \bar{c})$

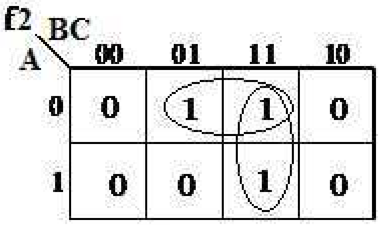
$f8 = \bar{b}\bar{c}(\bar{a} + a) + \bar{a}\bar{b}d$

$f8 = \bar{b}\bar{c} + \bar{a}\bar{b}d$

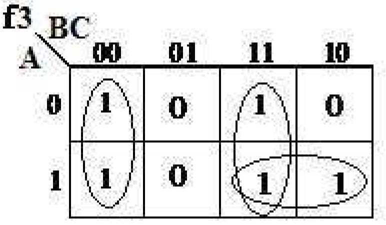
Exercice 4 : Simplifier par la méthode graphique (Karnaugh) les fonctions suivantes :



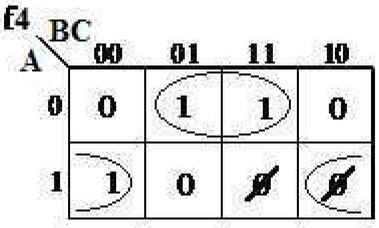
$F1 = A.\bar{B}.\bar{C} + \bar{A}.C$



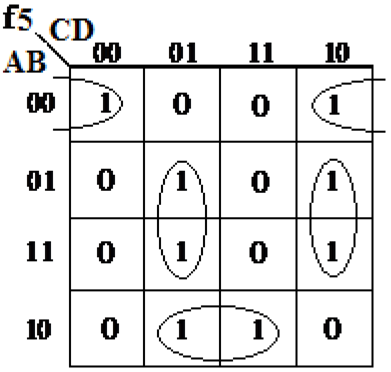
$F2 = \bar{A}.C + B.C$



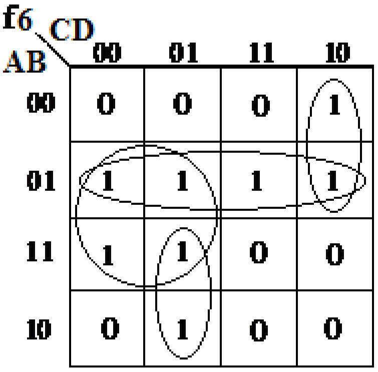
$F3 = \bar{B}.\bar{C} + B.C + A.B$



$F4 = A.\bar{C} + \bar{A}.C$



$f5 = \bar{A}\bar{B}\bar{D} + B\bar{C}D + BC\bar{D} + A\bar{B}D$

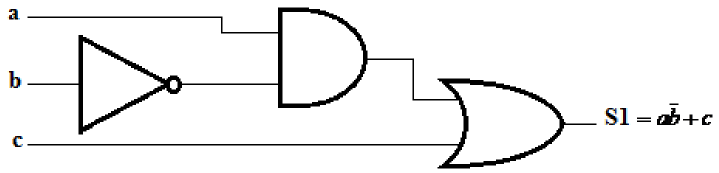


$f6 = \bar{A}B + B\bar{C} + A\bar{C}D + \bar{A}C\bar{D}$

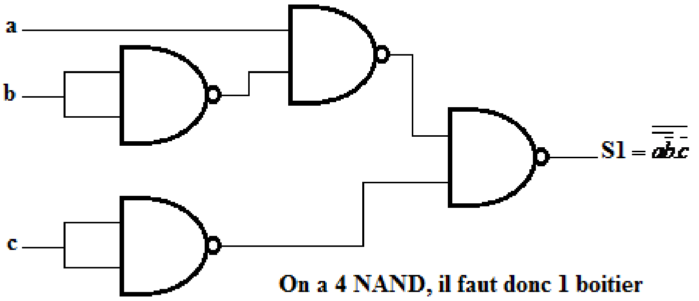
Exercice 5 :

$$S1 = a\bar{b} + c = \overline{\overline{a\bar{b} + c}}$$

$$S1 = \overline{\overline{a\bar{b}}.\bar{c}}$$



On a 1 NON, 1 AND, 1 OR
Il faut 3 boitiers (C. I)

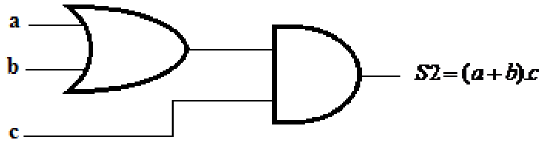


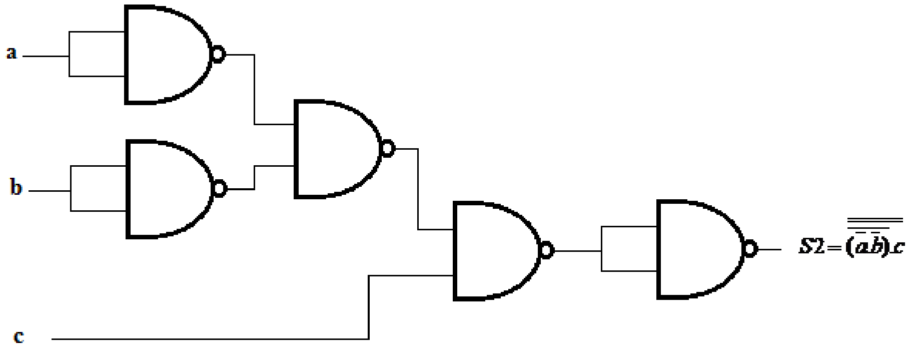
On a 4 NAND, il faut donc 1 boitier

$$S2 = (a+b).c = \overline{\overline{(a+b).c}}$$

$$S2 = \overline{(\bar{a}.\bar{b}) + \bar{c}} = \overline{(\bar{a}.\bar{b})}.\bar{\bar{c}}$$

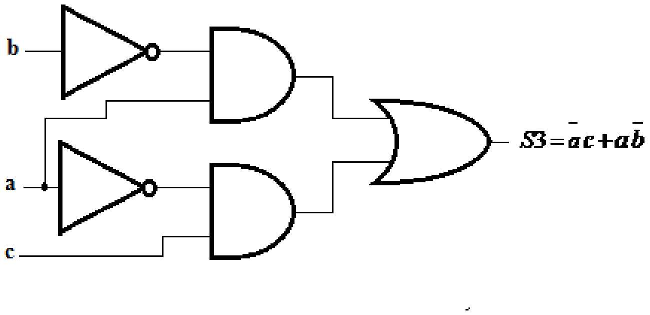
$$S2 = \overline{\overline{\overline{(\bar{a}.\bar{b})}.c}}$$

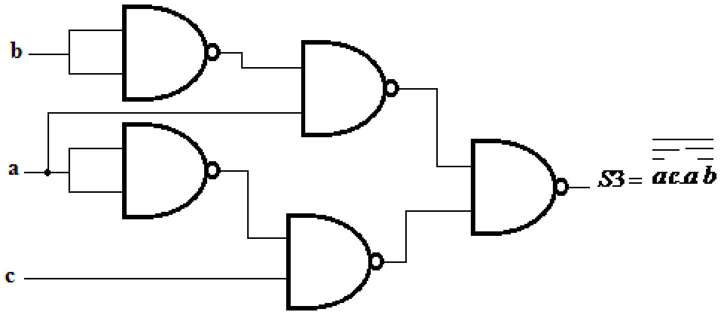




$$S3 = \bar{a}\,c + a\,\bar{b}$$

$$S3 = \overline{\overline{\bar{a}\,c + a\,\bar{b}}} = \overline{\overline{\bar{a}\,c}.\overline{a\,\bar{b}}}$$





Exercice 6 :

$$f = \overline{\bar{a}\,b + \overline{\mathrm{b\,c}}}$$

$$f = \overline{\bar{\mathrm{a}}\,\mathrm{b}}\,.\,\overline{\overline{\mathrm{bc}}}$$

$$f = (a + \bar{b}).b\,c$$

$$f = a\,b\,c + \bar{b}\,b\,c$$

$$f = a\,b\,c$$